Description and learning objectives

The goal of the project is to acquire specialized knowledge in graphics programming through project work. In the project you should demonstrate how techniques and theory from the course can be applied and/or expand on a field of your interest that is directly related to the course.

You should use a low-level graphics API and shaders in the project. You are encouraged to use OpenGL,but can alternatively use Vulkan, DirectX, OpenGL ES, WebGL or even your own rendering library (e.g. if you want to implement a ray-tracing algorithm). In some rare cases, if the project complexity requires it, you will be allowed to use Unity.

The project will be evaluated based on how graphics programming is being used in the project in terms of both performance and visual appearance. A minimum quality on the code is also expected.

It is possible and recommended to reuse the library from the exercises (ituGL). You can use any exercise as your starting point, depending on your project.

The project must be created individually.

Approval

You are required to send a brief project description to Chris to make sure that your project has a suitable scope and complexity.

Hand-in

The following must be handed in in the project:

- All source code and other resources (all that is needed to build and run your project)
- A project report:
  - The project report should give an overview of the project and describe how this is achieved and the theory that is used.
  - Between 4 and 8 pages of text, including small pictures. Longer does not mean better, just write what you think is relevant.
  - If you have used external resources, tutorials, source code, research paper or any knowledge in your project, this should be listed.
  - If you did not reuse the setup used for the exercises, then you should explain how to build your project.
  - One or more screenshots of the project, in a separate folder.

Hand-in deadline

The official ITU project hand-in deadline (Friday, 22th May 2026, 14:00)

Selecting a project idea

We recommend that you search for SIGGRAPH course resources (pages and videos), such as the annual course on Advances in Real-Time Rendering http://advances.realtimerendering.com/ or the 2010 course on Stylized Rendering http://stylized.realtimerendering.com/#info.

You can also propose a project on subjects that are only superficially discussed in the course, such as tessellation and geometry shaders, SDFs or ray-tracing.

Sample Project Ideas

Some possible ideas that are still available:

- Skinning: Deforming the mesh using bones, used for character animation
- Area lights
- Underwater scene: Fog, distortion, blue ambient light and other effects can be used
- Post processing effects: SSAO, SSR, FXAA (see lecture 9)
- GPU particles: Simulate particles using compute shaders (not possible in macOS)

Sample Project structure

You are free to organize your report as you want, but these are some of the things that it may contain:

- Motivation: Why you chose this project and not something else
- Problem statement: What you are trying to achieve
- Settings: If your project has options to play with, describe them here
- Implementation details: Describe how you implemented the solution, justifying your choices
- Additional features: Anything that you implemented and was not part of the requirements
- Future improvements: If you had extra time, how would you improve it?